

Read quality and full genome alignment after KRAS suppression

WEEK 2 PROGRESS REPORT

Yichuan Lin | University of Ottawa, Faculty of Science | 22 September 2026

Background and progress

RNA sequencing can reveal responses to KRAS small interfering RNA (siRNA) in pancreatic ductal adenocarcinoma. This week, read-quality assessment and 17 full-run genome alignments were completed.

Objective

Assess read quality, compare strand settings and summarize genome alignment.

Methods

Files were organized into 17 R1/R2 pairs to retain sample identities. Truncated imports were replaced with recovered files and checked by FastQC. FastQC 0.12.1 assessed quality and adapters in 34 files; MultiQC 1.35 combined 32 Galaxy reports and two verified local reports for comparison. Cutadapt was not run because base-quality and adapter checks passed; composition and duplication flags alone did not justify trimming. HISAT2 2.2.3 mapped paired reads to human hg38 to locate their genomic origins. Full runs used Phred+33, unstranded mode, no read limit, no end trimming and default alignment options. Mapping summaries recorded overall and concordant unique-pair alignment.

Full alignment results for 17 runs

Sample	Input pairs	Overall (%)	Unique (%)	Discordant (%)
Control M1	40,237,116	96.75	86.11	0.61
Control M2	45,764,716	96.78	85.19	0.58
Control M3	38,181,985	96.67	87.14	0.64
Control P1	45,946,458	96.73	87.25	0.57
Control P2	39,989,452	95.79	86.47	1.15
Control P3	47,840,900	96.41	87.03	0.54
Control P4	46,287,075	96.20	87.22	0.65
Control P5	45,023,981	96.75	88.32	0.57
KRAS M1	45,292,678	96.51	0.0030	75.56
KRAS M2	44,032,049	96.74	85.25	0.62
KRAS M3	42,961,283	96.64	86.95	0.60
KRAS P1	39,963,282	96.52	87.11	0.62
KRAS P2	42,632,214	96.49	87.50	0.69
KRAS P3	42,073,244	96.57	86.11	0.56
KRAS P4	40,452,536	96.77	88.32	0.52
KRAS P5	37,886,040	96.53	87.55	0.64
KRAS P5_v2	43,166,706	97.72	88.09	0.32

Table notes

Overall uses individual reads; unique and discordant use input pairs. Unique means concordantly aligned once. P5_v2 is SRR24828472; KRAS P5 is SRR24828471. Total: 727,731,715 pairs. Million-pair pilots are excluded. Overall alignment alone does not establish correct pairing.

Read quality results

All files contain 150-base reads. Composition and duplication checks flagged all 34 files; duplication ranged from 57.0% to 67.3%. Unequal transcript abundance can contribute to duplication in RNA libraries, so duplicates were not automatically removed. Both SRR24828471 mates had GC-distribution warnings, which do not establish contamination. Full-run overall alignment ranged from 95.79% to 97.72%.

Strand comparison and interpretation

Unstranded, forward-stranded (FR) and reverse-stranded (RF) pilots used the first million pairs from Control M1 and KRAS P1, not random read subsets. Overall alignment was 96.63% and 96.40%, respectively, under all three settings. Concordant unique alignment was 86.40% versus 86.44% (FR/RF) for Control M1, and 87.43% throughout for KRAS P1. These rates did not resolve strand specificity. RSeQC showed nearly equal orientation fractions in both samples, supporting unstranded mode; other runs were not individually tested. These different cell lines cannot isolate treatment effects. Gene counting and differential expression were outside this week's report.

References

1. Galaxy full-run summaries, pilots 122–136, MultiQC 165 and RSeQC outputs; 22 September 2026. Data: github.com/yichuanlin07/pdac-kras-rnaseq. Course slides: Week 2, Quality Control and Alignment.
2. HISAT2 manual. daehwankimlab.github.io/hisat2/manual/
3. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/
4. RSeQC documentation. rseqc.sourceforge.net/
5. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.

Author contributions

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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